KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2012
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss the ingredients of a cryptosystem briefly. (08)
   b) Discuss the characteristics of asymmetric cryptosystem. Make a comparison between block and stream ciphers. (10)
   c) Discuss the basic protocol of public-key algorithm so that it can be used for digital signatures. Does it satisfy the characteristics of signature? (10)
   d) Why and how one-way hash function is associated with digital signature? (07)

2. a) How RSA keys are generated and what are the limitations of RSA encryption algorithm? Explain with example. (10)
   b) Define 'primitive element' with example. How is it related to ElGamal cryptosystem? (10)
   c) What is meant by malleability? Show that RSA and ElGamal are malleable cryptosystem. (10)
   d) Explain triple DES mechanism. (05)

3. a) Define cryptographic hash function. Discuss some properties of it. (08)
   b) How ElGamal is a probabilistic cryptosystem? Explain. (08)
   c) Define symmetric key cryptosystem. Explain the steps of AES encryption algorithm. (12)
   d) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (07)

4. a) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (08)
   b) How is trade secret protection vanished by 'reverse engineering'? Explain. (08)
   c) Discuss the dimension of reliability and integrity of database. What are the requirements for database security? Explain them. (11)
   d) Define sensitive data. Discuss the steps of interaction among a user, a trusted front end and a DBMS. (08)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the parameters that can make a computer network vulnerable? Discuss them briefly. (10)
   b) What do you mean by security and computer security? "Security measures must balance between security and productivity"- Justify the statement. (10)
   c) Discuss the threats to electronic mail. (07)
   d) Define risk analysis. Discuss the steps of risk analysis. (08)

6. a) Discuss about risk exposure and risk leverage. (10)
   b) How can fence register be used to provide extra benefit in relocation method? What are the advantages of two pair base and bound register? (13)
   c) Discuss tagged architecture with pictorial view. (12)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How does buffer overflow affect code and data? Discuss elaborately. (15)
   c) What is computer virus? Differentiate virus and worm. How is virus attached? (10)

8. a) How does virus gain control? Discuss with pictorial view. Differentiate copyright, patents, and trade secret. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (10)
   c) Differentiate laws and ethics. "Computer crime is hard to define and hard to prosecute"- Justify the statement. (15)



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